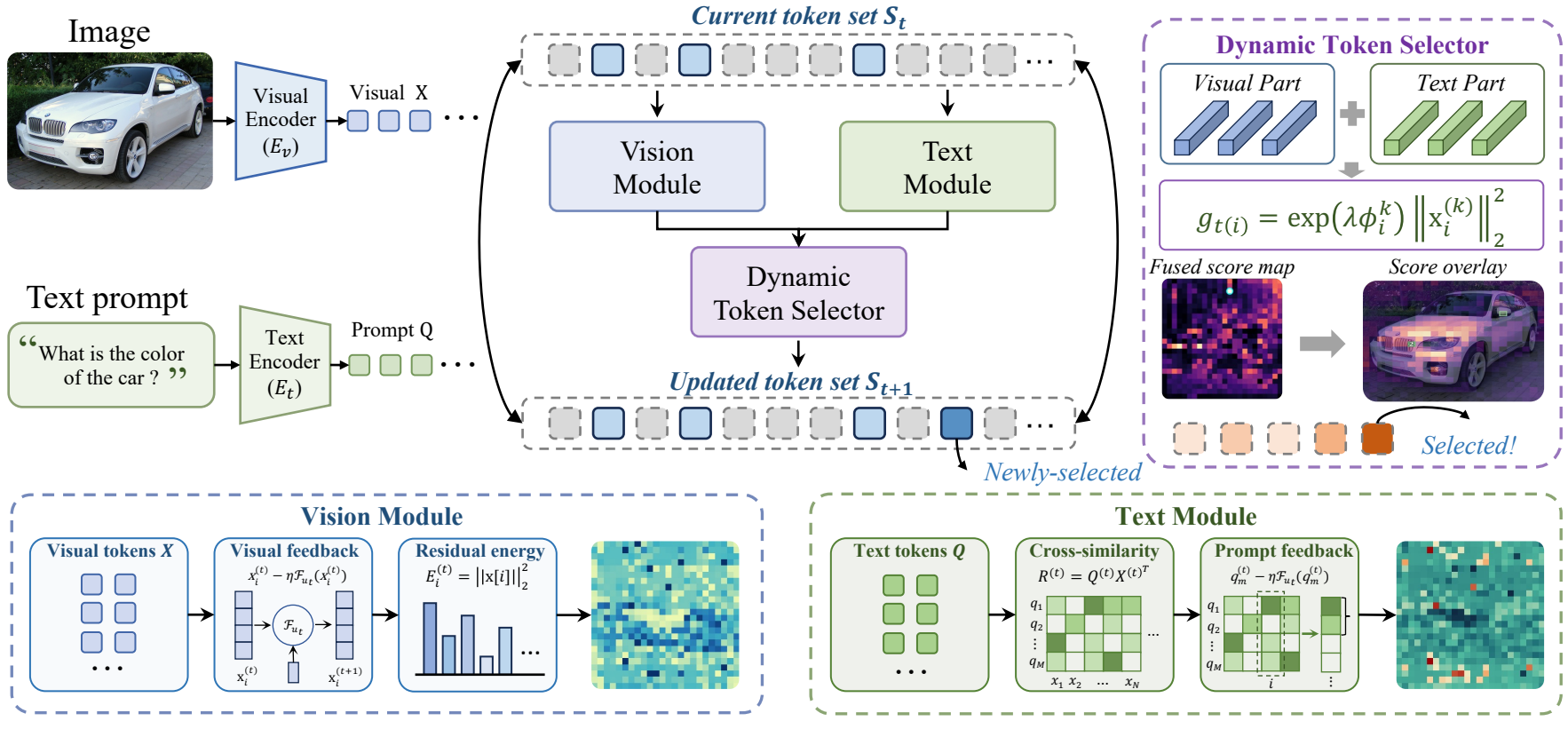

Image
Visual Encoder (E_v)
Visual X
Text prompt
"What is the color of the car ?"
Text Encoder (E_t)
Prompt Q
Current token set S_t
Vision Module
Text Module
Dynamic Token Selector
Updated token set S_{t+1}
Newly-selected
Dynamic Token Selector
Visual Part
Text Part
$g_{t(i)} = \exp(\lambda \phi_i^k) \left\| x_i^{(k)} \right\|_2^2$
Fused score map
Score overlay
Selected!
Vision Module
Visual tokens X
Visual feedback
$x_i^{(t)} - \eta \mathcal{F}_{u_t}(x_i^{(t)})$
Residual energy
$E_i^{(t)} = ||x[i]||_2^2$
Text Module
Text tokens Q
Cross-similarity
$R^{(t)} = Q^{(t)} X^{(t)^T}$
Prompt feedback
$q_m^{(t)} - \eta \mathcal{F}_{u_t}(q_m^{(t)})$